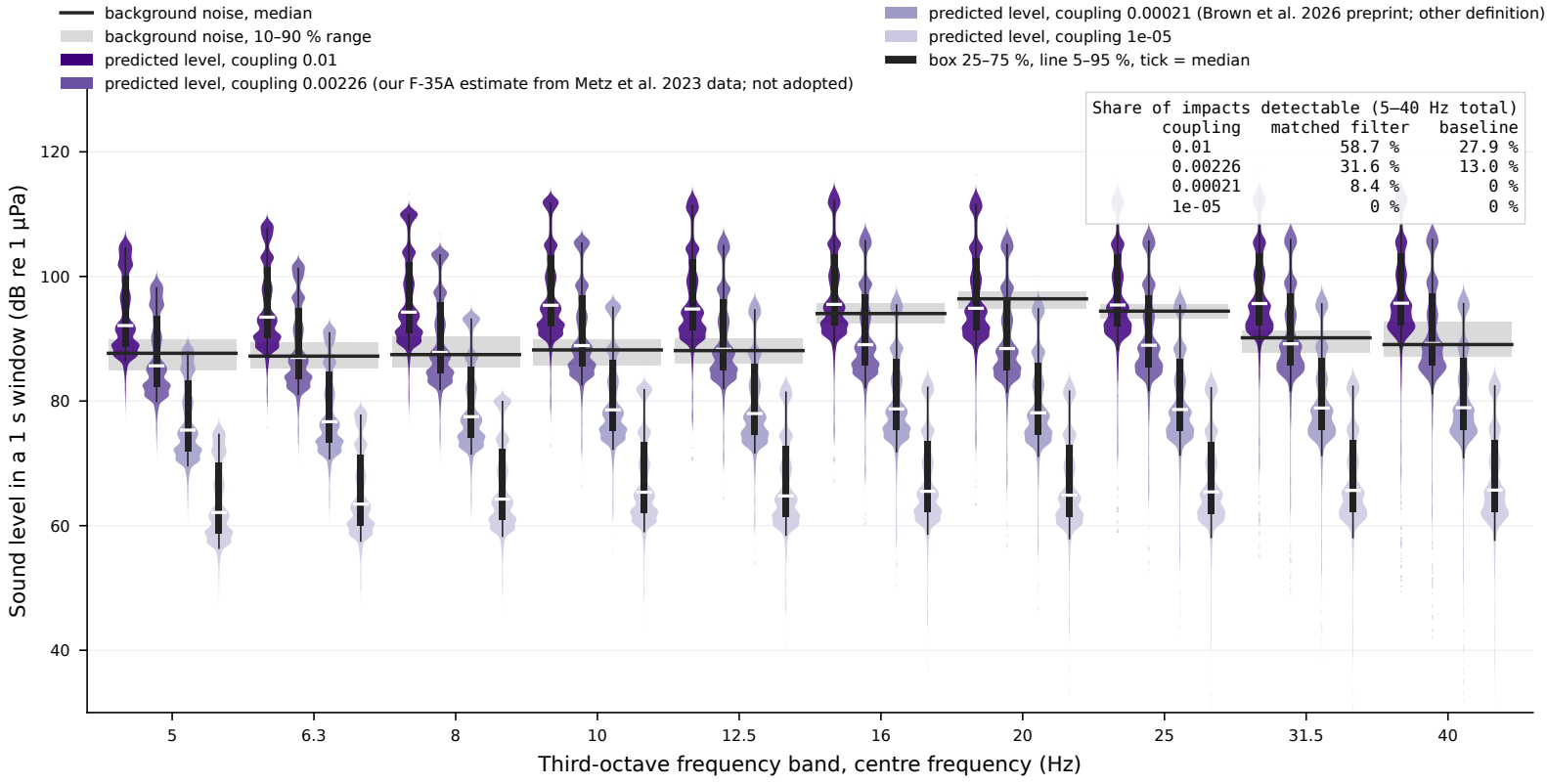
Cape Leeuwin (H01W): background noise against the predicted sound level of all simulated impacts, by frequency band



Plain note: The grey line is the median background noise at the station in each frequency band. The grey box holds 80 % of the noise levels. Each coloured shape shows all the simulated impacts: its width shows how many impacts give that sound level at the station. There is one shape for each of four assumed shares of impact energy that become sound. Where a shape is above the grey line, those impacts are louder than the noise in that band. The box at the top right gives the share of impacts that two detectors would find in the 5 to 40 Hz total. Below 5 Hz there is no prediction. These are estimates. They are not calibrated.

Technical note: Source posterior: end-of-flight run J (74 seeds, 313.6 M rows, four strata pooled by p_core/K) + northern supplement (impacts-ruleB.npz); weighted subsample 2 M draws per 00:19 option at dive probability p = 0.5 (run_j_levels.py, prereg 4277320). Panel: 00:19 option D0′ (held out), p = 0.5; Kish ESS of the subsample 1.90 M (Kish ESS of the full run J mixture for D0′ at p = 0.5: 1.34 M, run-j-windows/ess.json). E = contact kinetic energy (upper anchor), tau = tau90 clamped to 0.05–10 s, 10 m source, one-pole spectrum flat to 1/tau and f^-2 above; level = 1 s-equivalent third-octave band SPL (T_eff 2.68 s), dB re 1 uPa; eta only shifts the distribution by 10 log10 eta. Violins: weighted histograms of the 2 M draws (0.25 dB bins, 0.5 dB Gaussian smoothing), width normalised per violin; mass below 30 dB is not drawn. TL: whole-arc KRAKEN grid (offset-0 line, latitude-only, clamped 41–20 S); IMOS transmission loss: corrected RAM-along-the-arc table imos_tl_ram_arc.csv (imos_ram_grid.py, run_j_levels amendment 2; replaces imos_tl_ramcorr_arc.csv); IMS (H01W/H08S/H08N) loss unchanged. RAM delta at H01W/H08S, none at H08N; 3250 uncorrected. Noise: Blackman et al. (2004) App. B, May-June 2003, digitised (PSD x band width). Detectable = SNR of the 5–40 Hz total >= 7 dB (envelope matched filter, P_D 0.5, alpha 0.01) or >= 15 dB (STA/LTA baseline), model-matched upper bounds from the weak-signal trial on 3376 background; SNR against the module's 5–40 Hz noise medians (imos2b; IMS = 3376 proxy + Blackman offset, +4.8 dB triad gain). Noise is Blackman et al. 2004 (2003); 2014 cross-checks agree within about 1.5 dB (Royer 2016 spectrogram 101.7 dB after a raster-smoothing correction, Kadri 2024 trace 103.1 dB, against 101.7 dB). The earlier '2014 noise 6 dB quieter' case (Kadri spectrogram colours) is withdrawn. PREDICTIVE, NOT CALIBRATED. C1: propagation slope misfit +10.4 / +7.4 dB per octave at H01W / H08S (after the EW0112 source result a source-term caveat; air9 verdict NOT EXPLAINED, gap G23). C2: coupling efficiency eta is not calibrated and is treated as a prior range over at least four decades (bounds not yet ruled); 2.26e-3 is this project's own inversion of the F-35A arrivals at H11 digitised from Metz, Obana & Fukao (2023; the paper gives no energy, source level or coupling) with the 900 +/- 200 MJ energy from the Brown et al. (2026) arXiv preprint, the module TL (KRAKEN + RAM, not calibrated), source depth 10 m and an f^-2 spectrum (single-factor spread 3.4e-4 to 5.4e-2); one point inside the prior range, NOT adopted and not from an authoritative source. 2.1e-4 is the Arons-equivalent coupling in the Brown et al. (2026) preprint, a different definition. No detection is claimed.